

A Comparative Study of LSTM, Bi-LSTM, and BERT for Automated Essay Scoring

Khin Su Su Han
University of Technology (Yatanarpon Cyber City)
Pyin Oo Lwin, Myanmar
khinsusuhan2412@gmail.com

May Thu Myint
University of Information Technology
Yangon, Myanmar
maythumyint@uit.edu.mm

Abstract—The rapid progress in Artificial Intelligence (AI) and natural language processing (NLP) has created new possibilities for educational technology, particularly in automated essay scoring (AES). This research explores the potential of Artificial Intelligence (AI) in revolutionizing essay scoring within educational technology. Specifically, it focuses on comparing the effectiveness of three distinct approaches to automated essay scoring (AES): training a Long Short-Term Memory (LSTM) model, Bidirectional Long Short-Term Memory (Bi-LSTM) from scratch, and fine-tuning a pre-trained Bidirectional Encoder Representations from Transformers (BERT) model. Both models are trained and finetuned for each approach using the ASAP dataset from Kaggle's Learning Agency Lab - Automated Essay Scoring 2.0 competition. Their performance is evaluated based on metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), and Accuracy. To facilitate practical application, a Python web application built with Streamlit enables users to submit essays, choose a scoring model (LSTM, Bi-LSTM or BERT), and receive feedback based on the selected model's prediction. This study contributes valuable insights into the strengths and weaknesses of each approach, furthering our understanding of efficient model selection for AES tasks.

Keywords— *Automated Essay Scoring, LSTM, Bidirectional LSTM, BERT, NLP, ASAP Dataset, Model Fine-Tuning, Educational Technology*

I. INTRODUCTION

The integration of AI in education has revolutionized essay scoring, addressing the subjective, time-consuming, and difficult-to-scale nature of traditional grading. Automated Essay Scoring (AES) systems use Natural Language Processing (NLP) to evaluate essays efficiently and objectively, spurring significant research.

Advancements in AI have led to sophisticated models like Long Short-Term Memory (LSTM), Bidirectional Long Short-Term Memory (Bi-LSTM), and Bidirectional Encoder Representations from Transformers (BERT). LSTM and Bi-LSTM excel at processing sequential information, crucial for essay scoring, while BERT, with its transformer-based architecture, captures intricate text relationships through bidirectional analysis. This research compares these models' effectiveness in predicting essay scores using the Automated Student Assessment Prize (ASAP) dataset from recent Kaggle's Automated Essay Scoring 2.0 competition [5].

Recent studies highlight BERT's superior performance in various NLP tasks due to its understanding of complex word relationships, significantly improving scoring accuracy. Although LSTM and Bi-LSTM handle sequential data well, they generally lag behind transformer-based models like BERT in AES evaluations.

The study trains LSTM and Bi-LSTM models from scratch and fine-tunes a pre-trained BERT model on the

ASAP dataset. It evaluates their performance using metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), and Accuracy. Additionally, a Python web application developed with Streamlit enables users to submit essays, select a scoring model, and receive scores and feedback, providing practical tools for educators and institutions. This comparative analysis of LSTM, Bi-LSTM, and BERT models contributes to the development of more effective AES systems, enhancing the efficiency and reliability of essay assessments.

The structure of the paper is organized as follows: Section II reviews the literature review; Section III delves into the background theory; Section IV details the system implementation; Section V discusses the results and provides a comprehensive comparison of the three models; and Section VI addresses the limitations and proposes directions for future research. Finally, the conclusion is presented in Section VII.

II. LITERATURE REVIEW

The development of Automated Essay Assessment has been intricately connected to the progress in natural language processing (NLP) and machine learning. Initial AES models mainly depended on superficial features like word frequency, sentence length, and basic syntactic patterns to estimate essay scores. While these models were pioneering at the time, they often failed to grasp the subtle meaning and context within the text, resulting in inaccurate or inconsistent scoring. The introduction of Long Short-Term Memory (LSTM) networks significantly enhanced the handling of sequential data, making them a favored option for AES tasks due to their capability to model long-term dependencies and the evolution of ideas in essays [3]. LSTMs have been especially successful in capturing the flow and coherence of arguments, which are essential elements in holistic scoring.

The emergence of Bidirectional Encoder Representations from Transformers (BERT) brought further advancements in NLP by allowing models to comprehend context in a bidirectional manner. Unlike traditional models, BERT analyzes text in both directions, better capturing relationships between words in a sentence. This feature is vital for tasks like essay scoring, where the coherence and context of ideas greatly impact the final score [2]. BERT's outstanding performance in various NLP tasks, including text classification and sentiment analysis, has made it a prominent candidate for AES [8].

Recent studies have continued to investigate the application of machine learning models in Automated Essay Scoring (AES), aiming to enhance both accuracy and generalizability. Taghipour and Ng (2016) implemented LSTM networks to create an AES system that significantly improved scoring accuracy by effectively capturing the

sequential structure of text [3]. Their research emphasized the importance of understanding the logical progression of ideas in an essay, which is crucial for holistic scoring. Similarly, Devlin et al. (2019) introduced BERT, which has been widely used in text classification and has shown considerable potential in AES due to its deep contextual understanding [2].

In more recent research, hybrid models and data augmentation techniques have been explored to boost AES performance. For instance, a study by Butt et al. (2021) utilized transfer learning by fine-tuning pre-trained models like BERT for specific AES tasks, leading to improved generalization across various prompts and essay types [4]. A comprehensive survey by Gao et al. (2020) examined the state-of-the-art techniques in AES, identifying the limitations of existing models, such as their difficulty in handling diverse writing styles and biases in scoring [7].

The primary motivation for this comparative study stems from the desire to determine which approach is more effective for AES: training LSTM and Bi-LSTM models from scratch or fine-tuning a pre-trained BERT model using the ASAP dataset from Kaggle's Learning Agency Lab - Automated Essay Scoring 2.0 competition.[5] While BERT has demonstrated superior performance in many NLP benchmarks, it is computationally more expensive to train and deploy compared to LSTM and Bi-LSTM. Therefore, it's essential to assess whether the potential performance gains offered by BERT justify the increased computational overhead, particularly in educational settings where resources may be limited. By comparing the performance of LSTM, Bi-LSTM and fine-tuned BERT on the ASAP dataset, this research seeks to provide insights that can inform the development and deployment of more efficient and effective AES systems. The findings can help educators and institutions make informed decisions about which model to adopt based on their specific needs and constraints.

III. BACKGROUND THEORY

A. Deep Learning

Deep learning, a specialized area within machine learning, leverages artificial neural networks composed of numerous layers to capture intricate patterns within data. This approach has brought significant advancements across various domains, particularly in natural language processing (NLP), by automating the feature extraction process from raw data, which used to require manual effort. In the context of NLP, deep learning models, particularly those based on LSTM and BERT architectures, have been instrumental in advancing tasks such as machine translation, sentiment analysis, and automated essay scoring-

B. Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) networks are a type of recurrent neural network designed to address the vanishing gradient problem in traditional RNNs, making them highly effective for modeling sequential data. They excel in tasks where the order of information is critical, such as time series prediction and language modeling. LSTMs have played a significant role in machine translation, speech recognition, chatbots, automated writing assistance, and sentiment analysis by capturing long-range dependencies and monitoring word associations over extended sequences.

In the context of AES, LSTMs are used to model the sequential structure of essays, capturing the flow of ideas and the logical progression of arguments. The equations governing the LSTM cell during training are:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (1)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (2)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (3)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (4)$$

$$O_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (5)$$

$$h_t = O_t * \tanh(C_t) \quad (6)$$

Where f_t , i_t , and O_t are the forget, input, and output gates, respectively, C_t is the cell state, and h_t is the hidden state.[6]

C. Bidirectional Long Short-Term Memory (Bi-LSTM)

Bidirectional LSTM networks extend the capabilities of standard LSTMs by processing the input data in both forward and backward directions. This bidirectional approach enables the model to capture context from both past and future states, enhancing its understanding of the sequential data.

In the context of AES, Bi-LSTMs are used to model the sequential structure of essays more comprehensively by capturing the flow of ideas and the logical progression of arguments from both directions. The equations governing the forward and backward LSTM cells during training are similar to those of standard LSTMs, applied in both directions:

$$\text{Forward LSTM layer: } \vec{h}_t = \text{LSTM}(x_t, \vec{h}_{t-1}) \quad (7)$$

$$\text{Backward LSTM layer: } \overleftarrow{h}_t = \text{LSTM}(x_t, \overleftarrow{h}_{t-1}) \quad (8)$$

The final output h_t of the Bi-LSTM is obtained by concatenating the forward and backward hidden states:

$$h_t = [\vec{h}_t ; \overleftarrow{h}_t] \quad (9)$$

D. Bidirectional Encoder Representations from Transformers (BERT)

BERT is a transformer-based model that processes text bidirectionally, capturing the context of words by considering both preceding and succeeding words in a sentence. This approach enhances its ability to understand language nuances, making it highly effective in various NLP tasks, including essay scoring, where full context comprehension is essential. BERT excels in tasks like question answering, text classification, spam detection, sentiment analysis, and named entity recognition due to its superior understanding of contextual relationships.

In AES, BERT is employed to evaluate essays by understanding the context of sentences and paragraphs, allowing it to assess more subjective aspects like coherence, tone, and style [9].

BERT's architecture leverages the self-attention mechanism to capture dependencies between words, regardless of their distance from each other in the text [1]. The key equations used during the training of BERT models are based on the self-attention mechanism:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (10)$$

The final representation is obtained by concatenating the outputs of all attention heads and passing them through a linear transformation.

E. Evaluation Metrics

Evaluation of deep learning models typically involves metrics such as accuracy, precision, recall, F1 score for classification tasks, and Mean Absolute Error (MAE), Mean Squared Error (MSE), Accuracy and Quadratic Weighted Kappa (QWK) for regression tasks like AES. These metrics provide insight into the model's ability to generalize from training data to unseen data, ensuring that predictions are not only accurate but also meaningful in real-world applications.

In this comparative study, the performance of the three models is evaluated using the following metrics:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Accuracy

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (11)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (12)$$

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (13)$$

IV. SYSTEM IMPLEMENTATION

Fig. 1 shows the overview architecture of Automated Essay Scoring (AES).

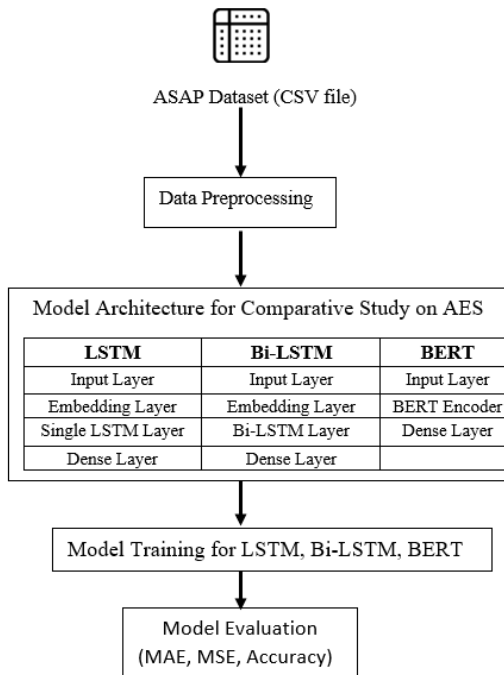


Fig. 1. Overview System Design of the model architecture

A. Dataset Description

This comparative study uses a dataset from the Learning Agency Lab Automated Essay Scoring 2.0 competition on Kaggle, which consists of approximately 24,000 student-written argumentative essays. These essays were contributed by students in grades 8 to 12 and were scored on a scale from 1 to 6 using a holistic scoring rubric.

The dataset is organized as three attributes which are the essay ids, the essays, and their corresponding scores, and 17308 rows in total, used as training data and fine-tuning for model development. These attributes are:

- `essay_id`: A unique identifier for each essay.
- `full_text`: The complete text of the student's essay.
- `score`: The holistic score assigned to the essay on a scale of 1 to 6.

TABLE I. EXAMPLE ROW OF THE DATASET

essay_id	full_text	score
000fe60	"I am a scientist at NASA that is discussing the "face" on mars....."	3

B. Data Preprocessing

The preprocessing of the ASAP dataset for both LSTM and Bi-LSTM, and BERT models involves several key steps to ensure the data is appropriately formatted for training and fine-tuning. First, the text data undergoes tokenization, where it is divided into individual tokens using model-specific tokenizers. Next, normalization is applied, converting the text to lowercase and removing special characters and punctuation to reduce noise. Since neural networks require fixed-length inputs, the text sequences are then adjusted through padding for shorter texts or truncation for longer ones, ensuring uniformity in input size. Additionally, the continuous essay scores are transformed into numerical labels through label encoding for regression tasks. Finally, the processed data is organized into batches, facilitating iterative weight updates during training via batch gradient descent.

C. Training the LSTM Model

The dataset for training was loaded from a CSV file on Google Drive. Essays were tokenized into integers based on a generated vocabulary and padded to a maximum length of 1000 tokens for uniform input size. An embedding layer with a dimension of 128 was used, initialized with random values. This model architecture included an embedding layer with an input dimension based on the tokenizer's word index, an output dimension of 128, and an input length of 1000 tokens. It featured a single LSTM layer with 64 units and a dense output layer with a 'linear' activation function. The model was compiled using the Adam optimizer and MSE as the loss function. It was trained over 10 epochs with a batch size of 64, using 80% of the dataset for training and 20% for validation. During training, the model aimed to minimize MSE and tracked MAE to assess performance. Real-time evaluation of the validation set allowed monitoring of the model's generalization capabilities.

D. Training the Bi-LSTM Model

This model was utilized for Automatic Essay Scoring (AES) due to its effectiveness in processing sequential data and understanding essay structure. It captures contextual dependencies from both past and future information, which is crucial for AES tasks.

It was trained using Word2Vec embeddings. The architecture consists of a Bidirectional LSTM layer followed by a regular LSTM layer. The dataset was split into 80% training and 20% validation sets. Text was tokenized and padded to ensure uniform input length. An embedding layer converted word indices into dense vectors using pre-trained Word2Vec embeddings, with an output dimension of 128 and an input length of 1000 tokens. This layer was set to non-trainable.

The Bi-LSTM layer had 64 units with a 'tanh' activation function, followed by a regular LSTM layer with 32 units. A Dense layer with one unit and a linear activation function provided the final prediction. The model was compiled with the Adam optimizer, using mean squared error as the loss function and mean absolute error as a metric. It was trained for 10 epochs with a batch size of 64 and a validation split of 0.2, enhancing its ability to capture dependencies and minimize prediction errors.

E. Fine-Tuning the BERT Model

The fine-tuning process of the pre-trained BERT model is conducted for automated essay scoring (AES). The dataset used for this task is sourced from Google Drive and contains essays along with their corresponding scores. Before feeding the data into this model, it undergoes a preprocessing step where each essay is tokenized using the AutoTokenizer from the Hugging Face library. The tokenizer converts the text into a format the model can process, including padding and truncation to ensure consistent input lengths. The essays are then split into training and test sets, with 80% of the data used for training and 20% reserved for testing. This split is crucial for evaluating the model's performance on unseen data. Additionally, the score column, representing the holistic scores of the essays, is converted into a numerical format and added as labels to the respective datasets.

The pre-trained model used for this task is specifically fine-tuned for sequence classification, with the output layer modified to predict a single score. This is achieved by using the AutoModelForSequenceClassification class from the Hugging Face Transformers library, initialized with the "bert-base-uncased" variant. The num_labels parameter is set to 1 since the task involves regression rather than classification. To fine-tune BERT for AES, the Trainer API from Hugging Face is employed. This API simplifies the training process by handling the optimization loop, evaluation, and model saving. The training arguments are carefully defined to ensure optimal performance. The model is trained for 3 epochs, balancing between sufficient training and avoiding overfitting.

V. RESULT AND DISCUSSION

In this comparative study, the LSTM and the Bi-LSTM models for Automated Essay Scoring (AES) were trained on Google Colab, utilizing the platform's computational

resources, specifically a CPU with 12.7 GB of system memory. During the training process, the system's memory usage fluctuated between 3 and 5 GB for both models. Each model was trained over 10 epochs, with each epoch taking between 281 and 327 seconds for LSTM to complete and 460 and 510 seconds for Bi-LSTM. Given the regression nature of these two models, they produced continuous scores. To align with the holistic scoring approach in the dataset, these continuous scores were rounded to the nearest integer value. This adjustment allowed for the output to more accurately reflect the scoring system used in the dataset, facilitating a direct comparison between the model's predictions and the actual scores.

The pre-trained BERT model for Automated Essay Scoring (AES) was fine-tuned on Google Colab, utilizing the platform's GPU resource, with 15.0 GB of system memory. During the fine-tuning process, the system's memory usage consistently ranged between 2 GB and 3 GB. The fine-tuning was conducted over 3 epochs, with the total runtime for the entire process being approximately 1 hour and 11 minutes. Each epoch contributed to refining the model's predictive capabilities, resulting in training losses that progressively decreased. Despite the regression nature of this model, which inherently produces continuous score predictions, these scores were rounded to the nearest integer values. This adjustment was necessary to align with the holistic scoring system utilized in the dataset, enabling a more accurate and meaningful comparison between the model's predictions and the actual scores provided in the dataset.

A. Trained LSTM and Bi-LSTM Model Evaluation



Fig. 2. Training Loss and Validation Loss for LSTM and Bi-LSTM over Epochs

Fig. 2 provides a comprehensive view of the performance and evolution of these two models, LSTM and Bi-LSTM during training over ten epochs. The first model's training loss significantly decreases from 2.9842 to 0.0524, indicating robust learning and improved accuracy on the training data. The second model also shows a notable decline in training loss, from 2.1316 to 0.3000, suggesting effective learning, albeit at a slightly slower rate than the first one. The validation loss for the first model fluctuates, decreasing from 1.0457 to 0.6560, reflecting improved generalization to unseen data but with some consistency challenges. The second model's validation loss steadily declines from 0.9314 to 0.4313, showcasing its better generalization across epochs. This figure highlights both

models' effectiveness in learning from the data and generalizing to new examples, with the second model showing slightly superior performance in terms of validation loss.

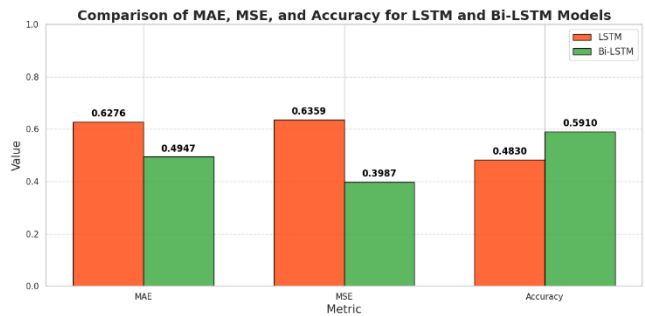


Fig. 3. MAE, MSE, and Accuracy for LSTM and Bi-LSTM Models

Fig. 3 presents a comparative analysis of the performance metrics between these two models. The first model, LSTM exhibits an MAE of 0.6276, indicating a relatively higher average absolute error in its predictions compared to the second model, Bi-LSTM, which achieved a significantly lower MAE of 0.4947. This suggests that the second model performs better in terms of prediction accuracy. The MSE for the first model is recorded at 0.6359, while the second model demonstrates an even more favorable performance with an MSE of 0.3987. The lower MSE of the second model indicates improved predictive reliability by minimizing the average squared differences between predicted and actual values. Regarding classification effectiveness, the first model's accuracy stands at 0.4830, whereas the second model achieves an accuracy of 0.5910, showcasing a substantial improvement in correctly predicted instances. This figure highlights the superior performance of the Bi-LSTM model across all metrics, offering valuable insights into its enhanced predictive capabilities over the standard LSTM model.

B. Finetuned BERT Model Evaluation

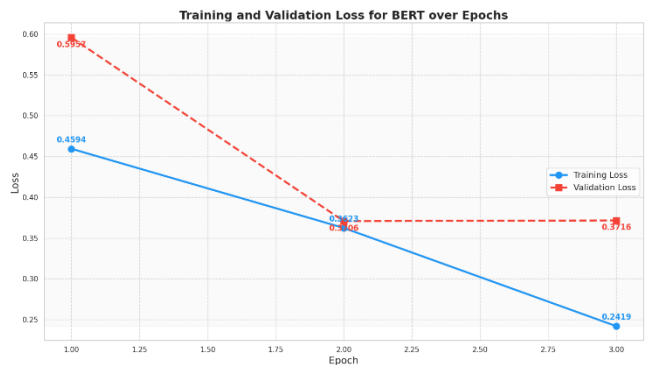


Fig. 4. Training Loss and Validation Loss for BERT over Epochs

Fig. 4 describes training loss and validation loss during the fine-tuning of the pre-trained BERT model for evaluating performance. Training loss, which measures how well the model fits the training data, consistently decreases over epochs. Initially, the training loss is 0.4594, dropping to 0.3623 in the second epoch and further to 0.2419 by the third epoch. This steady reduction indicates effective learning and

improved model performance on the training data. In contrast, validation loss reflects the model's performance on unseen data. Starting at 0.5957 in the first epoch, it decreases to 0.3706 in the second epoch but slightly increases to 0.3716 in the third epoch. This drop followed by a slight increase suggests the model is learning well but may be approaching a point of diminishing returns or potential overfitting. The observed trend shows that while the model fits the training data better, attention is needed to ensure it generalizes effectively to new data. This balance is crucial for optimizing model performance and avoiding overfitting.

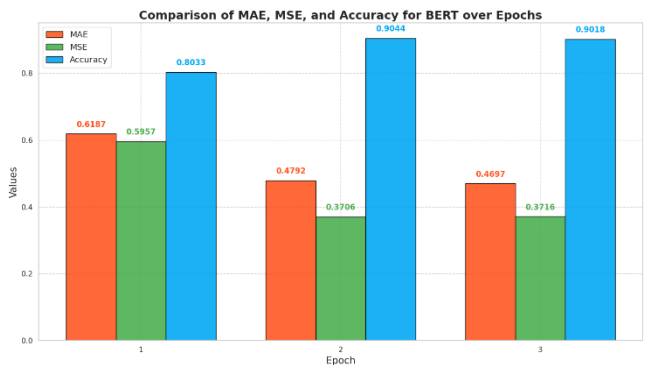


Fig. 5. MAE, MSE, and Accuracy for BERT over Epochs

In Fig. 5, the evaluation metrics highlight the BERT model's performance over three epochs. At Epoch 1, the MAE was 0.6187 and MSE was 0.5957, indicating higher prediction errors, with an Accuracy of 0.8033. By Epoch 2, the MAE decreased to 0.4792 and MSE to 0.3706, showing significant improvement, and Accuracy increased to 0.9044. In Epoch 3, MAE slightly improved to 0.4697, MSE stabilized at 0.3716, and Accuracy remained high at 0.9018. These metrics reveal consistent improvements and high Accuracy, demonstrating effective learning and adaptation.

C. Model Performance Discussion

The model performance evaluation metrics: MAE, MSE, and Accuracy with results are presented in the following table for clarity.

TABLE II. MODEL EVALUATION METRICS

Model	MAE	MSE	Accuracy
LSTM	0.6276	0.6359	0.4829
Bi-LSTM	0.4947	0.3987	0.5910
BERT	0.4729	0.3706	0.9044

The comparative study of the LSTM, Bi-LSTM, and BERT models for automated essay scoring provides significant insights into their respective capabilities and limitations. As illustrated in the evaluation bar chart, the performance metrics for each model in the following Fig. 6 are analyzed to understand their relative strengths and weaknesses.

In Fig. 6, BERT consistently outperforms both LSTM and Bi-LSTM models in automated essay scoring. BERT achieves the lowest MAE of 0.4792, indicating superior

accuracy in predicting essay scores compared to LSTM's 0.6276 and Bi-LSTM's 0.4947.

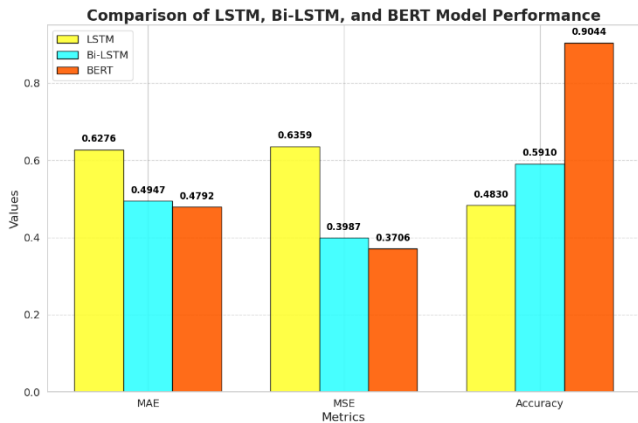


Fig. 6. Comparison of LSTM, Bi-LSTM, and BERT Model Performance

Similarly, BERT exhibits the lowest MSE of 0.3706, significantly lower than LSTM's 0.6359 and Bi-LSTM's 0.3987, highlighting its consistency and ability to minimize large deviations in predictions. Moreover, BERT's accuracy is remarkably higher at 0.9044, compared to LSTM's 0.4830 and Bi-LSTM's 0.5910, showcasing its enhanced capability in correctly categorizing essays into their respective score bins. The superior performance of BERT can be attributed to its transformer-based architecture, which leverages bidirectional attention mechanisms to capture complex contextual dependencies more effectively than the sequential processing approach of LSTM and Bi-LSTM. This comparative analysis underscores the significant advantages of fine-tuning transformer models like BERT over traditional sequential models such as LSTM and Bi-LSTM in achieving higher performance metrics for automated essay scoring.

VI. LIMITATIONS AND FUTURE RESEARCH

A. Dataset Constraints

The ASAP dataset from Kaggle's Learning Agency Lab - Automated Essay Scoring 2.0 competition provides scores based on a holistic evaluation approach. This presents a limitation for developing models that predict essay scores based on specific rubrics rather than holistic criteria. Additionally, the dataset's focus on argumentative essays may impact the model's performance when applied to other essay types, such as descriptive or narrative forms. To address these limitations, future research could explore creating or utilizing datasets with rubric-based scoring systems and including a variety of essay types to enable more generalized and adaptable scoring models.

B. Rubric-Specific Models

This dataset's reliance on holistic scores limits the development of models designed to predict essay scores based on specific rubrics. Future research should explore alternative approaches to address this limitation: fine-tuning large language models (e.g., GPT-4, Meta's LLaMA, Mistral's mode, and Google's Gemini) with rubric-specific

data for detailed and personalized feedback, adapting BERT variants (e.g., RoBERTa, DistilBERT) for rubric-specific scoring with lower resource demands, employing T5 (Text-to-Text Transfer Transformer) for versatile scoring and feedback generation, utilizing ALBERT (A Lite BERT) for efficient scoring with minimal computational resources, and leveraging XLNet for capturing long-range dependencies in essays to enhance scoring accuracy. Additionally, leveraging EFL-specific models trained on datasets annotated by professional EFL teachers can enhance accuracy for non-native essays. Incorporating diverse datasets and essay types, alongside EFL annotations, will improve scoring precision and model effectiveness, advancing automated essay scoring toward more accurate and rubric-specific evaluations.

VII. CONCLUSION

This research demonstrates the comparative performance of training LSTM, Bi-LSTM, and fine-tuning pre-trained BERT in automated essay scoring using the ASAP dataset from Kaggle's Learning Agency Lab - Automated Essay Scoring 2.0 competition. The findings indicate that BERT offers superior performance, showing strong results. The developed web application allows users to explore these two models, providing insights into the most effective model for AES tasks. Future research should focus on addressing the limitations identified and expanding the scope of model applications in educational technology.

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